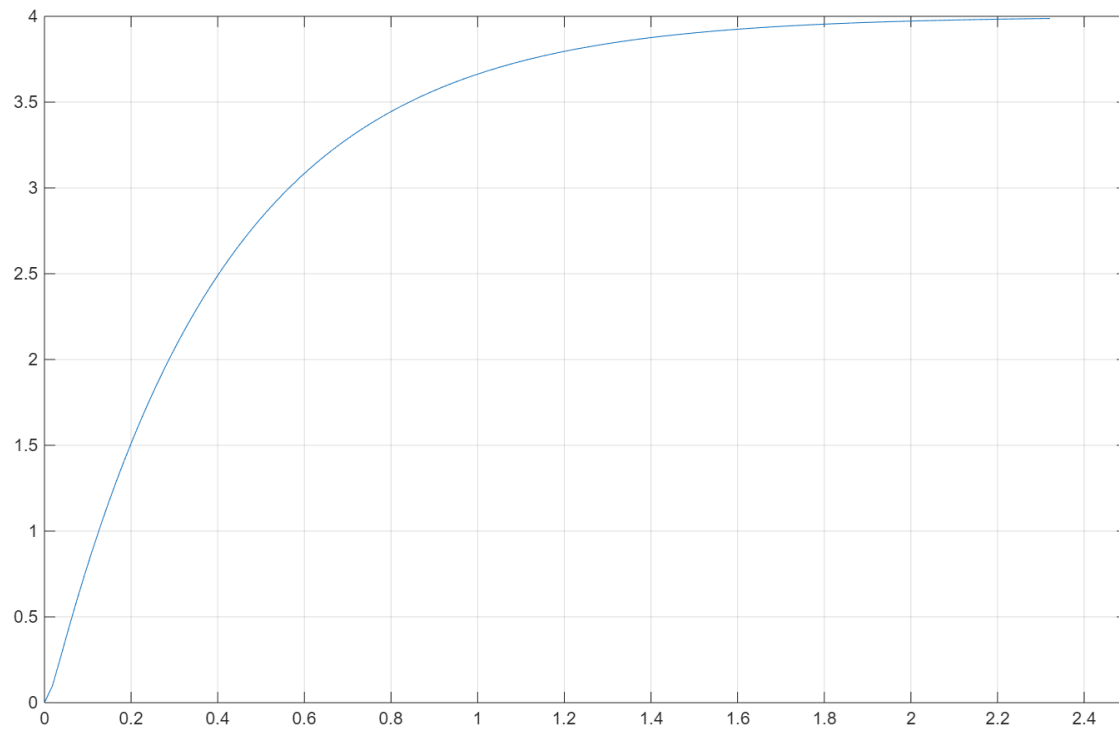


CT1 Q MEMO

[20 marks] You and some colleagues are working on identifying an unknown system. You start out by measuring the step response and get the following output:

Figure 1: A Step response of an unknown system



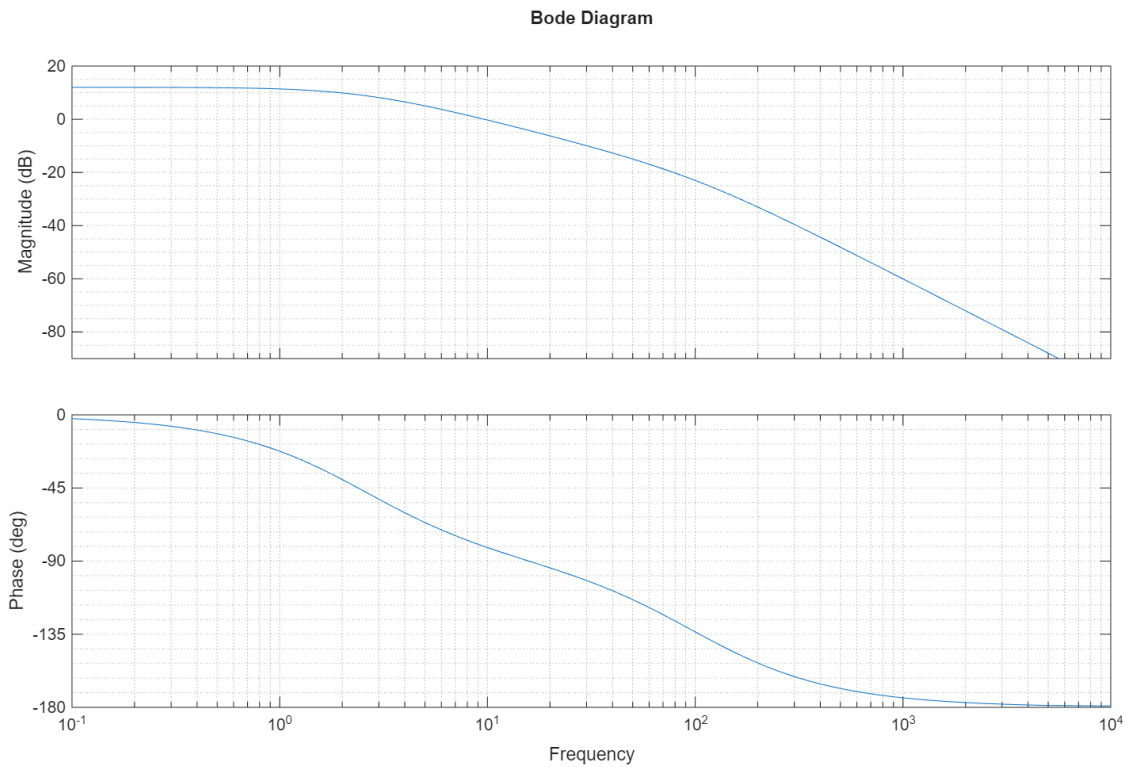
It is suggested that the transfer function of this system is

$$P(s) = \frac{10}{s + 2.5}$$

- (a) Based on only the step response, show that $P(s)$ is a reasonable model for the given system. [3 marks]

A colleague then comes up to you and gives you the following Bode plot of the system after having done frequency-based tests:

Figure 2: A Bode plot of that same system



- (b) On the same set of axes as Fig. 2 sketch the approximate frequency response of $P(s)$ [6 marks]
- (c) Based on the Bode plots, does the order of $P(s)$ match the order of the system it models? If not, what is the order of the system? Support your answer with 2 features from the bode plot. [4 marks]
- (d) With this new information from the bode plot, would you still consider the initial model $P(s)$ valid if the operating point of the system is:
- Around 2 rad/s [2 marks]
 - Around 200 rad/s [2 marks]
- (e) Propose a transfer function model for the system that matches the order seen on the Bode plot [3 marks]

Solution:

(Transfer function $L(s) = \frac{1000}{(s+2.5)(s+100)}$)

- (a) Using

$$G(s) = \frac{A}{\tau s + 1}$$

and looking at Fig. 1 we find that the final value is 4, and so $A = 4$. (1) Looking for τ , we find $0.63 * 4 \approx 2.5$. (1) The time the output reaches 2.5 is at $0.4s$, so $\tau = 0.4$. (1) Substituting into $G(s)$ gives

$$G(s) = \frac{4}{0.4s + 1} = \frac{10}{s + 2.5}$$

Figure 3: True solution to b)

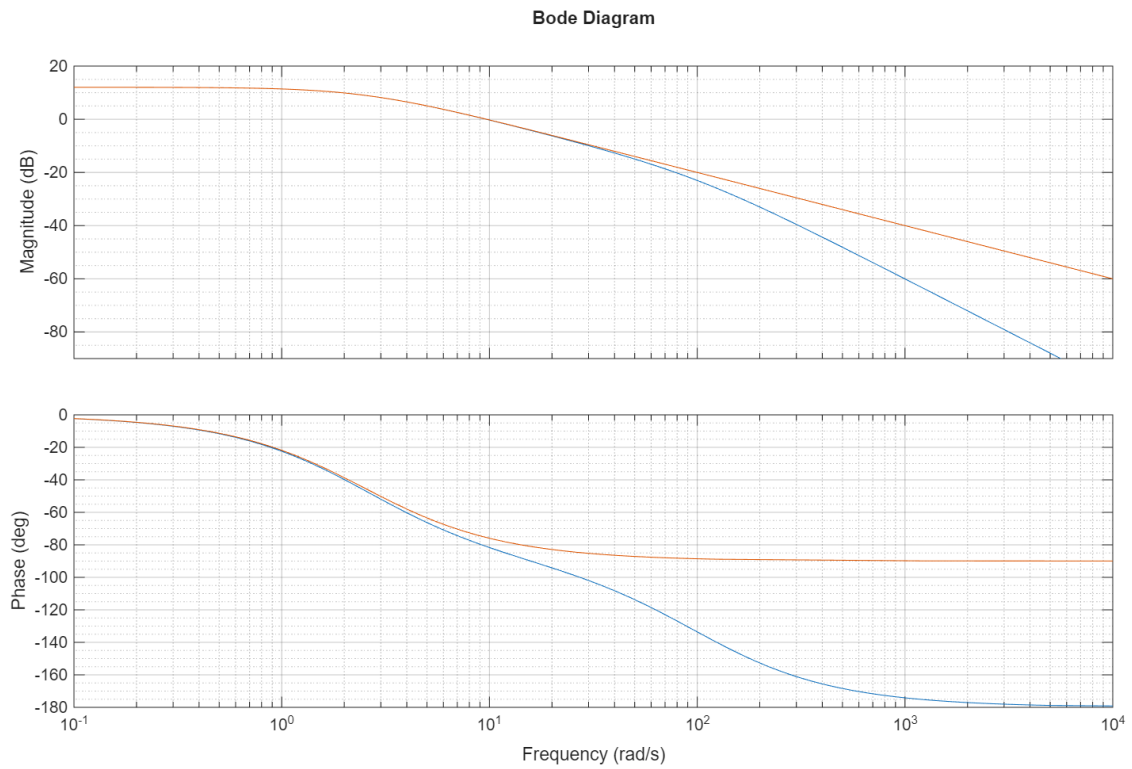
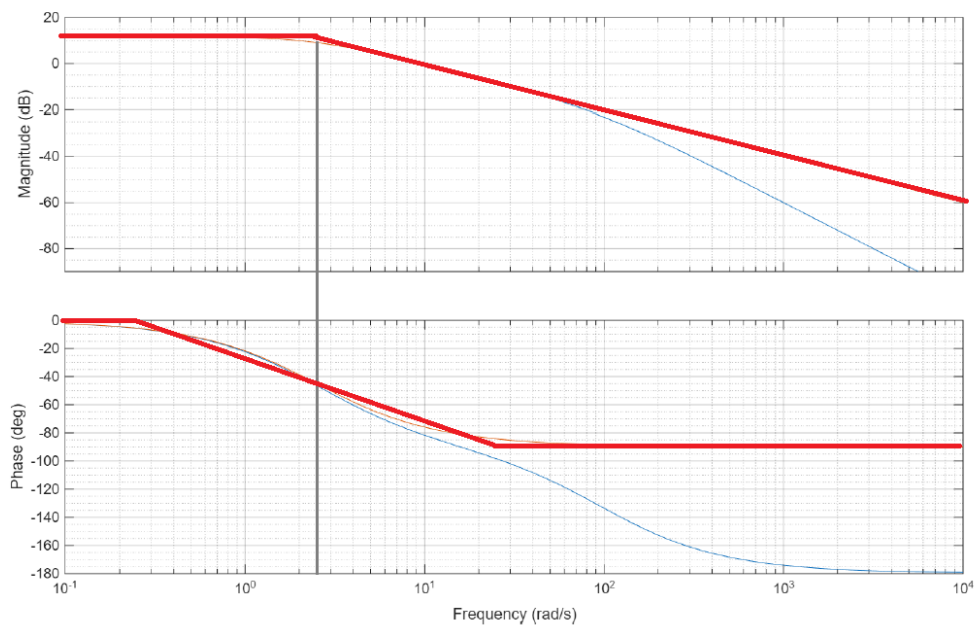


Figure 4: sketch solution to b)



- (b)
- (1) Low frequency gain of 12dB
 - (1) -20dB/dec asymptote at high freq
 - (1) Corner freq at 2.5 rad/s
 - (1) Phase starts at 0 and ends at

- (1) Phase of 45° at 2.5 rad/s
 - (1) Phase tends in a tan(ish, can be asymptote sketch) shape to -180°
- (c) No, we find both a -40dB/dec roll-off(1), and a high frequency phase of -180° (1) which tells us this is a second order system(1), while the proposed model $P(s)$ is only first order (1).
- (d) i) Around 2 rad/s the two plots are near identical both in phase and magnitude(1), so $P(s)$ will work for the plant.(1)
- ii) Around 200 rad/s the plots deviate both in phase and magnitude, and so $P(s)$ will not accurately represent the system. (1)
- (e) Using the given full bode plot we can see that the poles are significantly far apart(1), and so we can look for where the phase is at -135° to approximate the second pole location. We find this to be at around 100rad/s (1). With this we can add this pole and ensure we normalize the gain between proposed models:

$$L(s) = \frac{4}{\left(\frac{s}{2.5} + 1\right)\left(\frac{s}{100} + 1\right)} = \frac{1000}{(s + 2.5)(s + 100)} \quad (1)$$